MIL-D-51410(EA)
22 October 1974
(See section 6)

MILITARY SPECIFICATION

DYE, GREEN, ETHYL-BIS(2,4-DINITROPHENYL) ACETATE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of ethyl-bis(2,4-dinitrophenyl) acetate (EDA).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieve, Test.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6820

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D1193-72 - Reagent Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

Munsell Book of Color

(The Munsell Book of Color may be obtained from the Munsell Color Company, Inc., 2441 N. Calvert Street, Baltimore, Maryland 21218.)

Pharmacopeia of the United States of America, Eighteenth Revision

(Copies may be obtained from the Mack Publishing Company, Easton, Pennsylvania 18042.)

3. REQUIREMENTS

3.1 Chemical and physical characteristics. The dye shall conform to the chemical and physical characteristics of table I when tested as specified therein.

Table I. Chemical and physical characteristics

Characteristic	Requirement		Test paragraph
	Minimum	Maximum	
Molar absorptivity at 242 ± 2 nanometers (nm)	3.2×10^4	---	4.2.4.1
Melting point, °C	151	160	4.2.4.2
Volatile matter at 115° C, percent by weight	---	2.5	4.2.4.3
Particle size, percent by weight retained on:			4.2.4.4
149-micron (No. 100) sieve	---	2	
74-micron (No. 200) sieve	---	3	
44-micron (No. 325) sieve	---	10	

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3.2 Characteristics of paper containing dye. A paper approximately 6 mils thick and containing 1.0 ± 0.3 percent by weight of the dye shall be prepared from strong-bleached sulfite or sulfate pulp (see 6.5). The dye content of the paper shall be verified by testing the paper as specified in 4.2.4.5.

3.2.1 Color. The color of the paper shall match one of the Munsell colors listed in table II when tested as specified in 4.2.4.6.

Table II. Munsell colors for paper containing dye

Hue	Value	Chroma
5.0GY	9	1
10.0Y	9	1
5.0Y	9	1
N	9.5	
N	9	

3.2.2 Color stain produced by test reagent. When one drop of test reagent is applied to the paper as specified in 4.2.4.7, a visible color stain shall be produced within 40 seconds. The color of the stain shall match one of the Munsell colors listed in table III.

Table III. Munsell colors for stain produced by test reagent

Hue	Value	Chroma
10.0B	3	4
2.5B	3	4
7.5B	4	4
7.5B	3	4

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

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4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the dye produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of preparation for delivery. Sampling for examination of preparation for delivery shall be conducted in accordance with MIL-STD-105, inspection level S-4, using an AQL of 2.5 percent defective.

4.2.2.2 For test. Sampling for test shall be conducted in accordance with table IV. A representative specimen of approximately 50 grams (g) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

Table IV. Sampling for test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of preparation for delivery. The sample unit shall be one filled container, ready for shipment. Sample containers and the preparation for delivery thereof shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Marking incorrect, missing, or illegible

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4.2.3.2 For test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D1193-72 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Molar absorptivity. Weigh to the nearest 0.1 milligram (mg) approximately 0.40 g of specimen into a 500-milliliter (ml) volumetric flask. Add 250 ml of chloroform and stir to solution using a magnetic stirrer, if required. Remove the stirring bar, dilute to the volume mark with chloroform, and shake thoroughly to mix. Pipet 5 ml of the solution into a 500-ml volumetric flask, dilute to the volume mark with chloroform, and mix thoroughly. Using an ultraviolet lamp on a suitable double-beam recording spectrophotometer, 10-millimeter silica cells, and chloroform in the reference cell, run an absorbance curve of the solution from 200 to 300 nm. Run zero line in the same range using chloroform in both cells. Read the absorbance at 242 ± 2 nm. Calculate the molar absorptivity as follows:

$$\text{Molar absorptivity} = \frac{420A}{BC}$$

where: A = Blank-corrected absorbance at 242 ± 2 nm,
B = Cell path length in centimeters, and
C = Effective specimen concentration in grams per liter.

4.2.4.2 Melting point. Determine the melting point of the specimen in accordance with the Melting Range or Temperature test, Procedure for Class I contained in the Pharmacopeia of the United States of America, Eighteenth Revision.

4.2.4.3 Volatile matter. Weight to the nearest 0.01 g approximately 1 g of the specimen into a suitable moisture dish and weigh to the nearest milligram. Heat to constant weight at 115° ± 5° C, cool to room temperature in a desiccator, and weigh to the nearest milligram. Calculate the percent by weight volatile matter as follows:

$$\text{Percent volatile matter} = \frac{100(A - B)}{W}$$

where: A = Weight of dish and specimen before heating in grams,
B = Weight of dish and specimen after heating in grams, and
W = Weight of specimen in grams.

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4.2.4.4 Particle size. Use sieves conforming to RR-S-366. Nest the sieves specified in table I in order of decreasing mesh size (with the largest mesh sieve on top) on a receiving pan. Weigh to the nearest 0.1 g approximately 10 g of the specimen. Mix the specimen with sufficient water in a beaker by stirring with a glass rod to produce a smooth paste. Carefully wash the thoroughly wetted specimen through the sieves. (The use of a wetting agent is permitted). Dry the sieves in an oven at 70° to 75° C, cool to room temperature, and weigh the material retained on each sieve. Calculate the percent by weight material retained on each sieve.

4.2.4.5 Dye content of paper.

(a) Standard solution. Prepare a standard solution of the dye used to make the paper in accordance with the procedure used for the solution tested in 4.2.4.1

(b) Specimen solution. Cut a piece of the paper weighing approximately 1.0 g and weigh to the nearest 0.1 mg. Extract the dye with six 20- to 30-ml portions of chloroform. Quantitatively transfer each extract to a 250-ml volumetric flask. Dilute the combined extracts to the volume mark with chloroform and mix thoroughly. Pipet 25 ml of the solution into a 100-ml volumetric flask, dilute to the volume mark with chloroform, and mix thoroughly.

(c) Procedure. Run absorbance cures of the standard solution and of the specimen solution using the spectrophotometric procedure specified in 4.2.4.1. Read the absorbance of the standard solution and specimen solution at 242 ± 2 nm. Calculate the percent by weight dye in the paper as follows:

$$\text{Percent dye} = \frac{2AE}{BD}$$

where: A = Blank-corrected absorbance of specimen solution at 242 ± 2 nm,
B = Blank-corrected absorbance of standard solution at 242 ± 2 nm,
D = Weight of paper in grams, and
E = Weight of specimen used to make standard solution in grams.

4.2.4.6 Color of paper containing dye. Determine the Munsell color of the paper by comparison with a Munsell Book of Color.

4.2.4.7 Color stain produced by test reagent.

(a) Test reagent. Prepare a solution of 14.7 percent by weight tetrahydrofurfuryl alcohol, 3.4 percent by weight N-methylglucamine, and 81.9 percent by weight diethylene glycol. Stir for at least 30 minutes and allow to stand overnight. Centrifuge and use the clear liquid as test reagent.

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(b) Procedure. Place one drop (0.2 ml) of the test reagent on the paper containing the dye and measure the time required for the development of a stain. After 40 seconds, determine the Munsell color of the stain by comparison with a Munsell Book of Color.

5. PREPARATION FOR DELIVERY

5.1 Packing, level C. A quantity of 40.0 (+0.2 or -0) pounds of dye shall be packed in a nominal 5-gallon capacity steel pail furnished with a gasketed, fully-removable lug cover. The dye shall be packed in a manner to prevent alteration of contents and to assure safe delivery to destination without damage or leakage. The dye shall be packed to meet carrier acceptance at the lowest rates in compliance with requirements of carrier rules and regulations applicable to the mode of transportation.

5.2 Marking. Each pail shall be marked by any means which provides legibility as follows:

(a) Instructions on side of pail: FOR COVERED STORAGE ONLY
(b) Nomenclature as cited in the contract or purchase order shall be applied to the cover and side of each pail
(c) National Stock Number (NSN) or part number when NSN is not given
(d) Government contract or purchase order number
(e) Net weight of contents
(f) Lot number, date of manufacture, and date of pack.
(g) Contractor's name
(h) Additional markings as may be required by the contract and the contractor's policies and procedures
(i) Appropriate address markings

6. NOTES

6.1 Intended use. The dye covered by this specification in intended for use in the manufacture of M8 chemical agent detector paper.

6.2 Ordering data. Procurement documents should specify the title, number, and date of this specification.

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM E29).

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6.5 Manufacture of paper containing dye. A method of making paper hand sheets containing dye is described in "Production Engineering Study for M8 Paper Dyes," Report No. 11, August, 1969, available from Edgewood Arsenal. However, the Government makes no warranty that paper made in this manner will meet the requirements of this specification.

6.6 Supersession data. This specification supersedes and includes the requirements of Edgewood Arsenal Purchase Description 197-54-840, dated 4 October 1963.

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